

PATENT RESEARCH REPORT

1. ABSTRACT

This technical report synthesizes the methodology and experimental evidence from patent US7795344B2, focusing on the preparation, hydrogenation, and application of polymer nano-particles designed for elastomer modification, including Hydrogenated Nitrile Butadiene Rubber (HNBR) and related rubber matrices. The research scope defines the synthesis of monodisperse, nano-scale block copolymers featuring a crosslinked poly(alkenylbenzene) core—typically polystyrene—and a brush-like surface layer composed of conjugated dienes or polyalkylenes. The primary polymerization approach relies on living anionic dispersion polymerization in hydrocarbon solvents like hexane, initiated by organolithium compounds (such as n-butyllithium or functionalized lithium initiators). Following diblock copolymerization, micelle formation is induced, and the core is crosslinked using divinylbenzene (DVB) to yield stable nano-particles with a mean average diameter under 100 nm. A critical process trend highlighted is the subsequent catalytic hydrogenation of the outer conjugated diene layer. Using a homogeneous catalyst system prepared from nickel octoate and tributyl aluminum, the unsaturated diene units are hydrogenated to saturated polyalkylene blocks, achieving conversion levels of approximately 95%. The resulting hydrogenated nano-particles exhibit excellent dispersion and compatibility when incorporated into rubber compositions. Claim 2 explicitly identifies nitrile rubber (NBR) as a target matrix, where these nano-particles serve as reinforcing agents. The formulation trends demonstrate that the addition of these engineered nano-particles significantly enhances the tensile strength, modulus, and viscoelastic properties of the host elastomer compounds, making them highly suitable for demanding applications such as gaskets, engine mounts, and high-temperature rubber components.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7795344B2
- Patent Title: Nano-particle preparation and applications
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2010
- Relevance: The patent describes the synthesis of hydrogenated block copolymer nano-particles (with a crosslinked polystyrene core and a hydrogenated

polybutadiene/polyalkylene surface layer) and explicitly claims their incorporation into nitrile rubber (NBR) matrices to improve mechanical and viscoelastic properties.

Polymerization / Synthesis Method - Monomer 1: Butadiene (21.6 wt% in hexane blend) - Monomer 2: Styrene (33 wt% in hexane blend) - Solvent: Hexane (0.51 kg initial charge) - Initiator: n-Butyllithium (5.4 mL of 1.5 M solution in hexane) - Crosslinking Agent: Divinylbenzene (50 mL) - Polymerization Temperature: Value not specified. - Polymerization Conversion: About 100% - Hydrogenation Catalyst: Nickel octolate (111 mL of 8 wt% in hexane) and Tributyl aluminum (266.4 mL of 0.68 M in hexane) - Hydrogenation Conversion: About 95% based on H1 NMR analysis - Reaction Time: 2 hours for butadiene polymerization, 2 hours for styrene polymerization, 1.5 hours for crosslinking

Relevant Experimental Evidence - Example 8 describes the synthesis of hydrogenated nano-particles: a reactor was charged with 0.51 kg of hexane and 1.04 kg of a butadiene/hexane blend (21.6 wt% butadiene), initiated with 5.4 mL of 1.5 M butyllithium. After 2 hours, 0.68 kg of styrene/hexane blend (33 wt% styrene) was added. After another 2 hours, 1.8 kg of hexane and 50 mL of divinylbenzene (DVB) were added to crosslink the core, achieving ~100% conversion. - In Example 8, hydrogenation of the resulting nano-particles was performed in a 4 L reactor using a nickel catalyst solution prepared from 111 mL of nickel octolate (8 wt% in hexane), 37 mL of hexane, 0.6 mL of cyclohexene, and 266.4 mL of tributyl aluminum (0.68 M in hexane). The hydrogenation conversion of the butadiene units reached approximately 95% based on H1 NMR analysis. - Example 24 details the synthesis of functionalized nano-particles using 508 g of hexane, 1.04 kg of butadiene/hexane blend (21.6 wt% butadiene), and initiating with 6.3 mL of a 1.3 M solution of HMI-Pr-lithium in hexane, followed by addition of 680 g of styrene/hexane blend (33 wt% styrene) and crosslinking with 50 mL of DVB. - Example 32 demonstrates the hydrogenation of the butadiene-based nano-particles to a 95% conversion level (verified by H1 NMR) and subsequent compounding using a 50 g Brabender mixer. - Claim 2 explicitly targets the incorporation of these polymer nano-particles into nitrile rubber (NBR) and other elastomers to form modified rubber compositions with improved physical properties.

Technical Relevance This patent provides a highly detailed, reproducible recipe for synthesizing hydrogenated block copolymer nano-particles that are structurally compatible with nitrile rubber (NBR). The use of a nickel/aluminum-based hydrogenation catalyst to saturate the diene phase (achieving 95% conversion) is directly relevant to the chemistry of Hydrogenated NBR (HNBR), demonstrating how to achieve thermal and oxidative stability in the elastomer phase.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

In conclusion, the synthesis of hydrogenated block copolymer nano-particles as disclosed in US7795344B2 provides a viable methodology for reinforcing nitrile rubber (NBR) matrices. The process successfully utilizes living anionic polymerization of butadiene and styrene in hexane, followed by micellar crosslinking with divinylbenzene to form well-defined core-shell nano-particles. Saturated polyalkylene surface layers are subsequently achieved via a highly efficient nickel-octolate/tributyl-aluminum catalytic hydrogenation process, yielding up to 95% conversion. For optimal synthesis of modified Hydrogenated NBR systems, it is recommended to maintain strict control over the block copolymer micellization, utilize organolithium initiators for narrow polydispersity, and employ the disclosed transition-metal catalyst system to ensure high hydrogenation efficiency of the diene elastomer phase.

5. REFERENCES

- US7795344B2 | Nano-particle preparation and applications | Not disclosed in the available patent text. | US | 2010 | <https://patents.google.com/patent/US7795344B2/en>